

Neil Chulani

Work Experience

Apple - Kernel Architecture Engineer (June 2026 - Present)

- Working on the Core Drivers Architecture team leading the SystemExtensions project (userspace drivers within MacOS applications) and developing critical software within kernel driver systems and bootloaders

Apple - Battery Firmware Engineering Intern (Summer 2023, Summer 2024, Summer 2025)

- Working with embedded systems in Apple's Battery and Custom Silicon team to develop & deploy secure update and secure boot loader, write drivers for I2C communication, and implement other RTOS specific features on ARM platforms

Apple - Platform Architecture Security (Summer 2024)

- Developed new framework to provide secure signing and attestation for emerging chips

Clay-Wolkin Fellow (Fall 2025)

- Applied research in VLSI and SOC Design developing the CORE-V Wally RISC-V processor
- Contributing to RISC-V Instruction Set Architecture Manual with emphasis on Hypervisor and Cache Extensions

Johnson & Johnson Vision through Harvey Mudd Engineering Clinic Program (Spring 2025)

- Created an Ultra High Speed (UHS) camera and optical system for imaging cavitation bubbles produced by femtosecond laser and corneal eye tissue interactions during LASIK eye surgery

Arcadia Tractor - Software & Hardware Engineer (June 2021 - August 2022)

- Worked on an autonomous golf driving range ball picker and developed a Controller Area Network (CAN Bus) battery monitor system. Built golf ball detection machine learning model, as well as other embedded development with ROS

College

Pomona College, Computer Science and Applied Math, Double Major

GPA: 3.98, Magna Cum Laude

Honors: Phi Beta Kappa, Sigma Xi, Blair Nixon Memorial Award, Pomona College Scholar

Relevant Coursework:

- Computer Science: Machine Learning, Reinforcement Learning, Real Time Systems, Compiler Design, Usable Security and Privacy, Computer Systems with Lab, Programming Language Theory, Algorithms, Data Structures & Advanced Programming
- Engineering, through Harvey Mudd College: System on Chip Design, Microprocessor Systems Design and Application, Very Large Scale Integration (VLSI), Advanced Problems in Engineering (Neuroscience and Optics Research)
- Mathematics: Number Theory and Cryptography, Abstract Algebra: Groups and Rings, Differential Equations and Modeling, Real Analysis, Probability, Computational Statistics, Mathematics of Big Data, Vector Calculus, Linear Algebra

On Campus

Research Assistant in Mobile Robotics Lab, RL Control for Autonomous Robots, PCB Design for Autonomous Robots

Research Assistant in Harvey Mudd Biophotonics Lab, Multicore Fiber Bundles for Neuroscience

Teaching Assistant for Microprocessor Systems, Real Time Systems, Computer Systems, Data Structures

Aceing Autism Program Director - Helping children with Autism develop and build social connections through tennis

Captain of Pomona-Pitzer Varsity NCAA Men's Tennis Team

Skills

Programming Languages: C, C++, Python, Swift, ARM Assembly, Haskell, Java, Swift, Ruby, JavaScript

Hardware Development: SystemVerilog, Verilator, Spike, Synopsys Custom Compiler, KiCAD, Solidworks, Fusion

Embedded Systems: Keil uVision, Segger, Logic Analyzers, Oscilloscopes, I2C, SPI, FreeRTOS, SWD, JTAG, CAN Bus

Projects

Presented at Xplore workshop and co-authored Simulation in Robotics tutorial at ICRA 2026

2x Finalist and Presenter in the Embedded Security Challenge at CSAW (2024 & 2025)

Senior Thesis: Formal Proofs for Post Quantum Cryptography

Programmable 8-Bit Breadboard (and later PCB) Computer based on SAP-1 Architecture using 74LS Logic Gates

FPGA based Hardware Security Module (HSM) with AES hardware accelerator using SystemVerilog

Camera Sensor built using 1200 Phototransistors with hardware image compression (NeoObscura)

References

Adrian Alting-Mees, Firmware Manager for Battery Management at Apple - a_altingmees@apple.com

Elad Efrat, Platform Architecture Security Manager at Apple - elad@dexiosis.com

650-880-7731

Phone

1230 BRENTWOOD STREET, LOS ALTOS, CA 94024

Address

NEIL22@ME.COM

Email